Cosmology with peculiar velocities of galaxies in the nearby Universe

Proposal overview

The goal of COSMOPV is to investigate the accelerated nature of the expansion of our Universe with observations of galaxy density and peculiar velocity fields. We aim at measuring the growth rate of structures, quantity that can probe dark energy models and deviations from general relativity on cosmic scales.

Our team will use state-of-the-art galaxy and supernova datasets from the Dark Energy Spectroscopic Instrument (DESI) and the Zwicky Transient Facility (ZTF), producing the best density and peculiar velocity fields in the Northern sky.

COSMOPV is divided in three major components:

1. **Simulations:** producing large samples of synthetic versions of DESI and ZTF surveys;
2. **Systematics:** quantifying the impact of observational systematics in real and synthetic datasets;
3. **Cosmology:** obtaining cosmological constraints on dark energy and modified gravity theories.

The optimal and unique combination of DESI densities and DESI+ZTF velocities is expected to yield 8.3% precision on the growth rate at late cosmic times. Combined with other growth rate measurements, COSMOPV's results have the potential to reduce uncertainties on the gravitional index γ by a factor of two.

Our local team has the required expertise and has acquired international recognition in this field over recent years, and will work hand-in-hand with experts from DESI and ZTF collaborations.

The success of COSMOPV will critically depend on the workforce in our team required to merge efforts across datasets and collaborations. Therefore we request funds to hire one PhD student and one experienced postdoctoral researcher to achieve the objectives of COSMOPV.

COSMOPV's team members

Last Name	First name	Current position	Role and responsibilities in the project	Involvement (person.month)
Bautista	Julian	Professor	Scientific coordinator	43.2
Fouchez	Dominique	Researcher	Expert in supernova cosmology. All WPs	4.8
Racine	Benjamin	Researcher	Expert in supernova cosmology. All WPs	4.8
Feinstein	Fabrice	Professor	Expert in supernova cosmology. WP2	19.2
Mena-Fernandez	Juan	Postdoc	Expert in clustering, simulations and lensing. WP1, WP3	14.4
Hanser	Corentin	Postdoc	Expert in CMB and peculiar velocities. WP1, WP2	14.4
Kebadian	Rafael	PhD student	Expert in peculiar velocities with SNIa. WP1 and WP2.	14.4
To be recruited		**PhD student**	WP1 and WP2	36
To be recruited		**Postdoc**	All WPs	36

I. Proposal's context, positioning and objective(s)

a. Objectives and research hypotheses

The accelerated nature of the expansion of our Universe remains one of the most mysterious puzzles in Physics, well consolidated by several recent and independent cosmological observations [1–5]. Explanations for such an acceleration require Physics beyond our standard model and/or ambitious modifications of our current theory of gravity: general relativity.

Dark energy

One of the potential solutions is to introduce a new exotic component of our Universe: **dark energy**. Dark energy has negative relativistic pressure, therefore acting as a source of anti-gravity, repelling galaxies on large cosmic scales. The simplest form of dark energy is equivalent to Einstein's cosmological constant Λ. The cosmological constant corresponds to a fluid with an equation of state $w = -1$ (relation between relativistic pressure and energy density), constant with time. Current observations require that at least 68% of our Universe's energy budget is in form of dark energy. Such a model is dubbed ΛCDM, where CDM stands for cold dark matter. More exotic dark energy models predict different values for w, with the only constrain that $w < -1/3$ to produce an accelerated expansion. A generalised model considers the equation of state of dark energy to be an time-evolving quantity [6, 7], which can be written as $w(a) = w_0 + w_a(1 - a)$, where a is the scale factor of the Universe ($a = 1$ today and $a = 0$ at the Big Bang) and w_0, w_a two fixed parameters. Such a model is referred to as w_0w_aCDM. In 2024, a combination of several cosmological measurements (see below) showed strong preference for an w_0w_aCDM Universe over a ΛCDM one by more than 3σ [5], showing hints of new Physics for the first time since the discovery of the acceleration in the 2000's.

Modified gravity

An alternative solution to explain the accelerated expansion of the Universe is to propose alternatives to general relativity (GR) to describe gravity on large cosmic scales. Those are referred to as **modified gravity theories**. Modified gravity effects have to manifest only on cosmic scales where the expansion is observed, since GR is compatible with extremely high-precision measurements made on Earth [8–12] in the Solar System [13, 14], on stars [15, 16], on binary and triple systems of pulsars [17–20], as well as in mergers of black holes and neutron stars [21, 22]. Direct imaging of black holes also are consistent with GR, in a very strong field regime [23]. There are several families of models trying to extend or modify GR to account for the observed acceleration of the Universe [24–28]. However, such models predict the same *expansion history* as dark energy, so we need an alternative observable to discriminate between dark energy and modified gravity models. One of the best observables is the *growth history* of the cosmic density field of the Universe and its impact on its velocity field.

Expansion rate and distance indicators

The expansion rate of the Universe, $H(z)$, as a function of time or redshift z, has been historically been measured via the average distance-redshift relationship of extragalactic objects. Redshifts are precisely measured with spectroscopy, but distances are a more challenging endeavour. Several distance indicators have been used in recent years: the Tully-Fisher (TF, [29]) and Fundamental Plane (FP, [30]) relations, as well as type-Ia supernovae (SNIa, [31]). The first two indicators (TF and FP) rely on empirical relations between galaxy properties and their luminosity or size, so in principle they can be applied large samples of galaxies. However those relations have a large intrinsic scatter of about 20-30% in distances compared to SNIa (5-10%). SNIa are more precise distance indicators but are less numerous than TF and FP galaxies The discovery of the accelerated expansion in the 2000's used samples of tens of SNIa [32, 33] while today's samples reach several thousands (see below). **COSMOPV will employ state-of-the-art samples of distance indicators to constrain models of dark energy and modified gravity.**

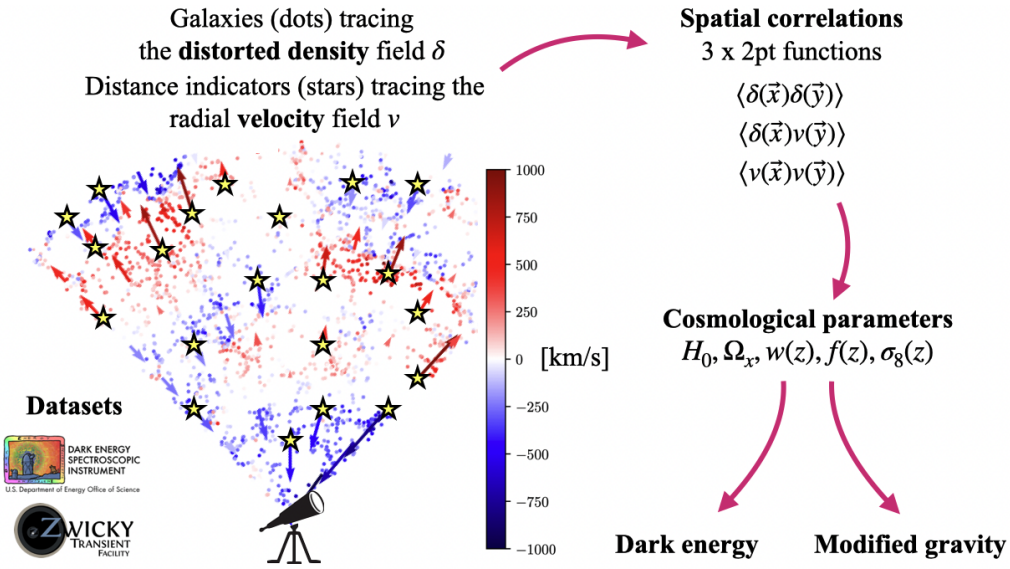

Figure 1. Schematic view of the COSMOPV. We will use state-of-the-art samples of galaxy redshifts and distance indicators to measure both the distorted density field and the radial peculiar velocity field to derive better constraints on the growth rate of structures, dark energy and modified gravity theories.

Growth rate of structures and peculiar velocities

The growth rate of structures $f(z)$ quantifies how fast initial perturbations of the cosmic density field, $\delta(\vec{x})$, grow with time or decreasing redshift z. The r.m.s. of the cosmic density field smoothed on scales of $8h^{-1}$ Mpc noted $\sigma_8(z)$ also quantifies structure growth versus time. The product $f(z)\sigma_8(z)$ can be measured via the impact of *peculiar velocities* of galaxies on the observed structures. Peculiar velocities (PV) are simply the motions of galaxies due to gravitational attraction from dense structures, therefore they are important to test gravity on cosmic scales. PVs modify redshifts due to the cosmic expansion. Both are generally undistiguishable, creating distortions in the observed galaxy density field, known as *redshift-space distortions* (RSD). The two-point statistics of the galaxy density field, $\langle\delta(\vec{x})\delta(\vec{y})\rangle$, becomes anisotropic w.r.t. the line of sight, allowing measurements of the growth rate $f(z)$ with spectroscopic galaxy surveys [34–36]. Moreover, if distance indicators are available for a large sample of galaxies, it is possible to break the degeneracy between velocity due to cosmic expansion and velocity due to proper motion. Measuring the velocity field $v(\vec{x})$ brings extra information than just measuring the distorted galaxy density field $\delta(\vec{x})$. The two-point statistics of the velocity field, $\langle v(\vec{x})v(\vec{y})\rangle$, also depends on $f(z)\sigma_8(z)$. Measuring three combinations of two-point functions, i.e., density-density, velocity-velocity and their cross correlation $\langle\delta(\vec{x})v(\vec{y})\rangle$, yield tighter constraints than just one of such statistics [37–40]. Figure 1 shows a schematic view of such a combination. **COSMOPV will employ state-of-the-art samples of distance indicators (FP, TF, and SNIa) to measure peculiar velocities and obtain the tightest constraints on the growth rate in the nearby Universe.**

b. Position of the project in relation to the state of the art

COSMOPV is timely because it will profit from state-of-the-art datasets to produce robust measurements of the growth rate of structures. Our team members are active members of the Dark Energy Spectroscopic Instrument (DESI) and the Zwicky Transient Facility (ZTF), and have the required expertise to achieve COSMOPV's goals.

The Dark Energy Spectroscopic Instrument (DESI)

Currently the best spectroscopic survey of galaxies, DESI will measure 63 million galaxy redshifts after eight years of observations (2021-2028) over 17,000 deg^2 [41]. DESI observes galaxies and quasars, which serve as excellent tracers of the distorted matter density field. By analyzing these redshift-space distortions (RSD), we can infer the growth rate of cosmic structures. The growth rate measurements from DESI Data Release 1 (DR1) [42] were published in 2024 [36] as well as their cosmological implications [43, 44]. Such growth rate measurements are displayed as red points in Figure 2 at $z > 0.1$.

DESI Peculiar Velocity Survey and recent results

The DESI Peculiar Velocity survey is a secondary program that will produce, when completed, distance measurements to 53k spiral galaxies from the Tully-Fisher relation and 133k ellipticals from the Fundamental Plane [45–47]. Those distances will yield a well sampled velocity field at $z < 0.1$, perfectly overlapping with the DESI Bright Galaxy sample [48], which provides an excellent density field.

Since 2024, the scientific coordinator of COSMOPV was nominated as co-chair of the DESI Transients and Low-z cosmology working group, responsible for the Peculiar Velocity survey. Since then, we are actively involved in the organisation and execution of the growth rate measurement with such data. In December 2025, we released a batch of seven publications describing the growth rate measurements with DESI DR1. We report a 12.5% measurement of $f(z)\sigma_8(z)$ at $z = 0.07$ [49–54].

The scientific coordinator was first author of the publication describing the simulated datasets [49]. Those simulations represent the largest and most accurate set for peculiar velocity studies so far, including 675 realisations of DESI DR1 galaxy and peculiar velocity samples. It was based on the AbacusSummit suite of n-body simulations [55] and an evolving halo occupation distribution model [56]. **In COSMOPV, we will build upon this expertise to produce larger and better samples of simulated data (Work Package 1).**

We were also lead authors of the growth rate measurement with the momentum power spectrum method [52]. This method is one of the three used with the same data, besides the maximum likelihood estimator [57] and correlation functions [54]. In those analyses, some unexplained systematic effects were found. First, the Tully-Fisher data at $0.05 < z < 0.10$ shows spurious correlations, so they were excluded from our baseline analysis. Second, residual spurious correlations persist in the final catalog, primarily affecting small scales, which motivated the application of a scale cut to the momentum auto-power spectrum. **Within COSMOPV, we will address these systematic effects by investigating their origin, correcting them when possible, and quantifying their impact on the growth rate of structures (Work Package 2).**

The Zwicky Transient Facility (ZTF)

ZTF is a photometric survey with large field of view and high cadence, ideal for the discovery of transient events, including SNIa [58]. Covering the whole sky down to declination of -30° and perfectly overlapping with DESI observations, ZTF discovered more than 30k SNIa over redshifts $0 < z < 0.2$, which will be used for cosmological studies. More than 5k SNIa were spectroscopically classified by ZTF or other telescopes. ZTF observations ran from 2017 to 2023, and now the ZTF collaboration, including members of COSMOPV, are working on an improved pipeline optimized for precision cosmology and dark energy studies, including the measurement of the growth rate from SNIa peculiar velocities. The ZTF Data Release 2 [59] contains half of the observations and is the last public release of ZTF data.

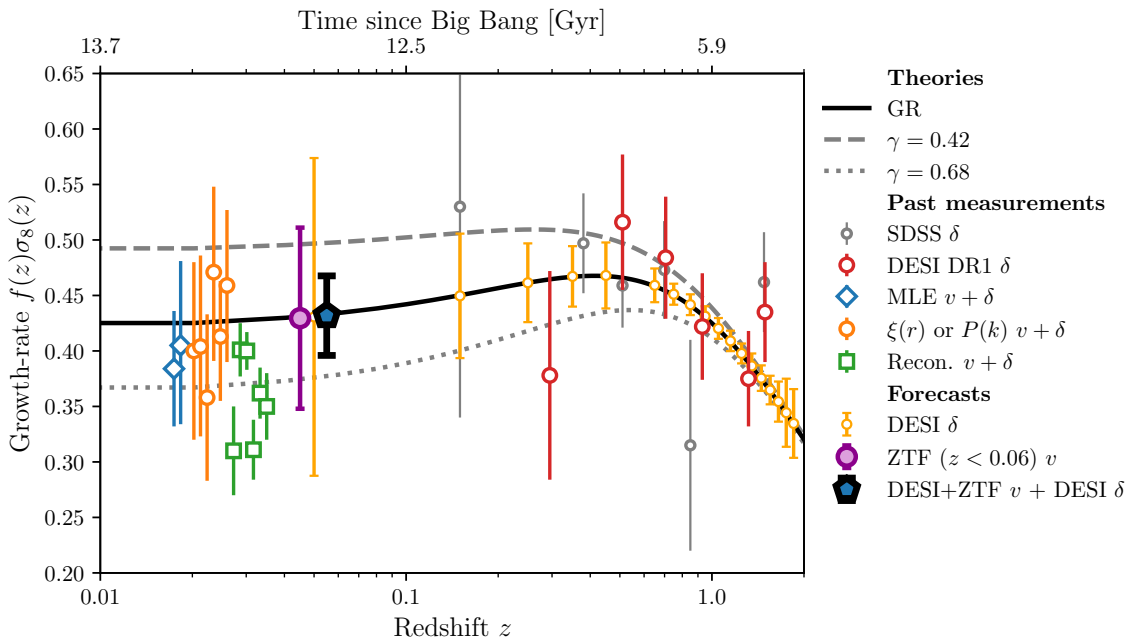

Figure 2. Growth rate of structures versus redshift from past measurements compared to expected results from COSMOPV, which will employ low-redshift galaxies and peculiar velocities from the Dark Energy Spectroscopic Instrument (DESI) and the Zwicky Transient Facility (ZTF).

ZTF peculiar velocities and recent results

Preliminary work on simulations started at CPPM to quantify the potential for cosmological measurements with ZTF SNIa sample. We developed an analysis pipeline for 1) simulating realistic observations of ZTF SNIa, a package called SNSIM[1] and 2) measuring the growth rate of structures with ZTF SNIa peculiar velocities using the maximum likelihood estimator [60]. This pioneering work in ZTF predicts an uncertainty of 17% on the growth rate from SNIa peculiar velocities only, without a galaxy density field (Figure 2).

We generalized this pipeline to include cross correlations with a distorted galaxy density field, correctly accounting for non-linear clustering and wide angle effects. This new framework is called Field Level Inference Package (FLIP, [61]) and is publicly available[2]. Thanks to several numerical optimizations as well as the implementation of emulators, FLIP was used to quantify the potential for growth rate measurements with the Vera Rubin Observatory [62], which contains one order of magnitude more SNIa.

COSMOPV will build upon this expertise with SNIa peculiar velocities and apply it to the cosmology sample of ZTF SNIa. We will produce simulations of ZTF with improved realism (Work Package 1) and use them to quantify biases and systematic errors in real data (Work Package 2).

While ZTF DR2 concerns the spectroscopically confirmed sample of SNIa, there is also a huge potential to increase numbers and redshift range by employing a photometrically-classified sample of SNIa. This sample is 4 to 5 times larger than the spectroscopically-classified sample, and reaches redshifts up to $z < 0.2$. WP2 of COSMOPV also will explore this opportunity to improve growth rate measurements.

Combining DESI and ZTF peculiar velocities

COSMOPV will focus in the DESI and ZTF overlap, a perfect opportunity for cross-correlations and systematic cross-checks, and produce a joint measurement of the growth rate of structures. Based on our own Fisher forecasts and analysis of simulations using FLIP [60, 61], **COSMOPV has the potential of producing a 8.3% precision**

[1] https://snsim.readthedocs.io
[2] https://flip.readthedocs.io

growth rate measurement when combining DESI galaxies and DESI+ZTF velocities (Figure 2), representing the most precise and robust estimate at late cosmic times. This number is to be compared to the current 12.5% measurement of DESI DR1 PV only [52, 54, 57].

Cosmological implications of such a measurement will be explored in Work Package 3, including the standard ΛCDM model as well as extensions parametrizing modified gravity theories (or departures from GR).

c. Methodology and risk management

COSMOPV is organised in three connected work packages: simulations, systematics, and cosmology. Figure 3 shows the timeline of these three WPs, their tasks and corresponding actors.

Work Package 1: Simulations

WP1 will focus on the production and analyses of realistic synthetic datasets reproducing properties of real data from DESI and ZTF. Simulations are a key ingredient of any robust cosmological analysis and have several uses: testing our methodologies for biases, understand our uncertainties, compute covariance matrices of observables, and study the impact of systematic effects on cosmological results. This work package will push the state of the art in the production of synthetic datasets for both DESI and ZTF, increasing the number of realisations, as well as the realism of implemented observational features.

Task 1.1: Production of synthetic samples for the DESI and ZTF PV samples

Based on acquired experience with mock production for the DESI Peculiar Velocity survey [49] and the ZTF SNIa sample [60], we will produce larger samples of realistic mock catalogues simultaneously including both surveys and their respective observational properties.

We will use already available suites of n-body simulations such as AbacusSummit [55] or Uchuu [63, 64], which model precisely the density and velocity field to non-linear scales.

While precise, n-body simulations are limited in volume and the number of possible realizations we can obtain from them ($\mathcal{O}(100)$). To achieve larger number of realisations ($\mathcal{O}(1000)$), we plan to also use fast approximate simulations such as the EZMOCKS [65, 66] or GLAM mocks [67, 68]. Larger number of realizations are important to obtain precise covariance matrices and to test biases and systematic effects to higher statitical precision.

We will use current methods to imprint Tully-Fisher and Fundamental plane samples in DESI galaxy mocks as well as realistic ZTF light-curves, including peculiar velocities. By construction, those mocks will contain the cross-correlations between surveys, which will be correctly accounted for when deriving cosmological constraints from a joint analysis. Those new mock catalogues will reach higher redshifts than current sets, and will include redshift evolution of the clustering signal and galaxy properties as well as selection effects.

Task 1.2: Cross-covariance between methodologies and cosmological probes

Several methodologies are able to exploit a distorted density field and a peculiar velocity field (e.g. [40, 61]). Those are commonly applied to the same initial dataset, so results are correlated. For example, DESI DR1 PV growth rate measurements from three methodologies [52, 54, 57] applied to the same dataset have correlations estimated around 27 and 38% thanks to mock catalogs [49].

Using large sets of mock catalogues, we will precisely estimate the cross-correlation between methodologies applied to the same data, which is key to derive robust consensus results and their uncertainties (e.g. [69–71]). Using larger datasets such as DESI DR2 and DR3 and ZTF DR3 and DR4, we will investigate the cross-correlation between those methods to a higher statistical precision.

The same mocks will be also used to evaluate the cross-covariance between growth measurements from peculiar velocities and weak gravitational lensing, since both probes use information from the same overlapping

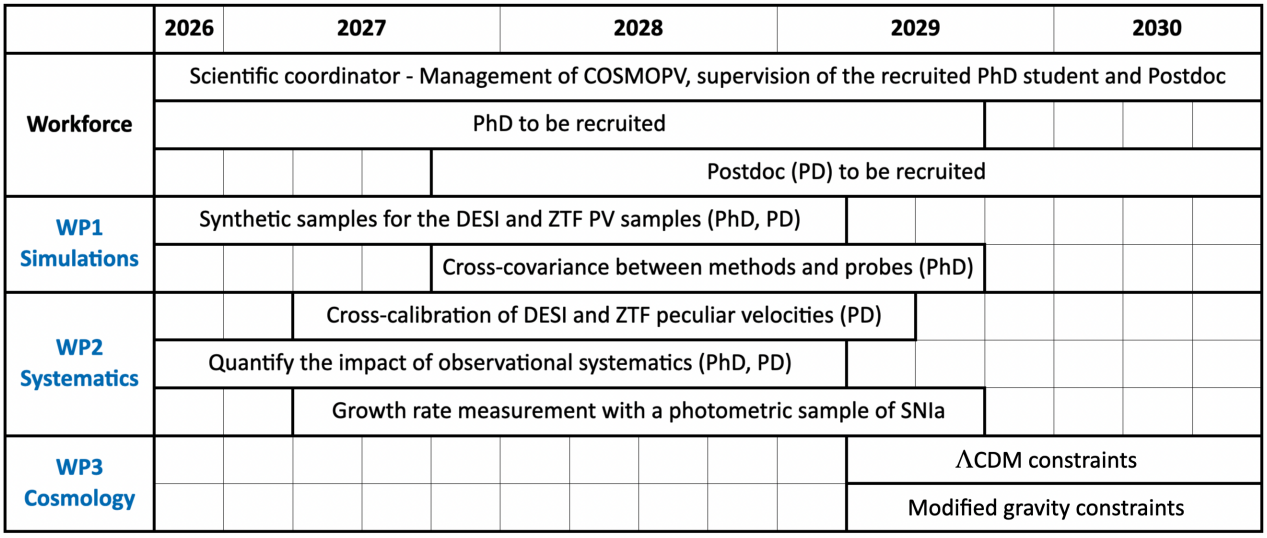

Figure 3. Gantt chart of the COSMOPV project.

volumes. The mocks will have to include ray-tracing to include the shear of background galaxies caused by the same matter density field used in peculiar velocity measurements.

Deliverables We will deliver large sets of precise and realistic simulations for both the DESI PV survey and the ZTF SNIa sample. First, they will be released internally for both collaborations so all members can use them. Second, they will be made available to the community for public use, after publication of the final results.

Risk management The production of mock catalogues is a time-consuming process and requires available computational resources. We expect to have access to computing time as our simulations are a key ingredient for both collaborations. For instance the DESI collaboration provides computing time at the NERSC supercomputer while ZTF provides access to the CNRS Computing Center. We also have available computing ressources at CPPM, which can also be used for this WP. Therefore we qualify the risk of this WP to be low.

Work Package 2: Systematic effects in real data

Focusing on an interplay between real data and simulations from WP1, WP2 will investigate potential systematic effects affecting our measurents with DESI and ZTF peculiar velocities. We will tackle systematics both at the level of creating a joint PV catalog after cross-calibrating both surveys, and at the level of growth rate measurements from those PV catalogs, looking for potential contamination by unwanted correlations in caused by observational artifacts.

Task 2.1: Cross-calibration of DESI and ZTF peculiar velocities

DESI will produce peculiar velocities from samples of distances obtained from Tully-Fisher and Fundamental Plane relations, while ZTF will do the same with type-Ia supernovae. Currently, those distance indicators are calibrated independently, using overlaps with existing sources with known distances [51, 59].

The first challenge for combining DESI and ZTF peculiar velocities is to perform a careful cross-calibration of both sets of distance indicators, making sure to understand any potential mismatches between them. Note that calibrating distance indicators also means providing a measurement for the Hubble constant H_0, even though so far such a measurement is mainly driven by the first two rungs of the distance ladder [51].

This task will deliver a joint catalog of peculiar velocities for DESI and ZTF, which will be used to derive a combined growth rate measurement.

Task 2.2: Quantify the impact of observational systematics

With the mocks from WP1, we will quantify for the first time the impact of several observational systematics to an unprecedented level of accuracy. Several effects can create spurious correlations in the observed velocity field, potentially biasing our growth rate measurements [72].

Peculiar velocities are derived from the residuals of the average distance-redshift relation. Such a relation is also measurement from the data, which suffers from selection effects (also referred to as Malmquist bias), i.e., at large distances fainter galaxies or SNIa are missing due to instrumental limitations or quality cuts. This problem is a key challenge in the study of the expansion of the Universe using SNIa, but it also affects Tully-Fisher and Fundamental plane distances in a similar way.

We will pay particular attention to the angular dependence of the Malmquist bias, the effect of dust attenuation, flux calibration residuals, among others. Preliminary work already started at CPPM, finding that a dipolar mis-calibration in the sky of about 0.02 magnitudes is sufficient to bias the growth rate measurement from ZTF SNIa [Kebadian et al. in prep].

Strategies will be put in place to mitigate such impacts when possible, otherwise a systematic uncertainty budget will be propagated to final results. We will use mock datasets from WP1, including those systematic effects to the best of our knowledge. Our final measurement will present both a statistical and a systematic uncertainty, making it a robust result compared to past measurements in the literature (see Figure 2).

Task 2.3: Growth rate measurement with a photometric sample of SNIa

This photometrically-classified sample can potentially increase the number of SNIa by factors of 5 to 6 compared to the spectroscopically-classified sample, also increasing the redshift range to $0 < z < 0.2$. Preliminary forecasts made at CPPM showed that the growth rate measurement could have uncertainties reduced by a factor of 2, relative to the spectroscopic sample.

Thanks to our simulations from WP1, we will be able to tackle new challenges brought by the use of a photometrically-classified sample of type-Ia supernovae. Building up on our experience with synthetic datasets of Rubin LSST SNIa [62, 73], we will extend this work to the final release of ZTF SNIa data (DR4). We will investigate to further detail the impact of classifiers of supernovae based on light-curves, contamination by non-Ia supernovae, and selection effects.

This work already started within the ZTF collaboration, and we would like to contribute to it thanks to the several outputs from COSMOPV. Given its ambitious nature, this task will have a lower priority compared to the previous ones.

Deliverables This WP will deliver a well-calibrated joint catalog of peculiar velocities from DESI and ZTF, which will be shared among participants of both collaborations. We plan to deliver as well a systematic error budget to the final measurement of the growth rate of structures from such a catalog.

Risk management Scientifically, this WP is **low risk** and we foresee no major blocking elements to achieve its objectives. Collaboration-wise, this WP requires an official agreement between DESI and ZTF collaborations. The idea will be to add ZTF members interested in the project to joint DESI as external collaborators. A well-justified proposal has to be written and needs to be accepted by the DESI board. There is a small risk that, if rejected, COSMOPV will have to rely on public data releases or unofficial agreements between DESI and ZTF members (post data releases). We believe the risk of rejection for such an agreement is **low**. The scientific coordinator was one of the authors of an accepted proposal inviting a few ZTF members as DESI external collaborators to participate in the production of joint simulations of DESI+ZTF for peculiar velocities, and a subsequent publication describing growth rate measurements [Ravoux, Bautista et al. in prep.].

Work Package 3: Cosmological implications of our measurements

WP3 will derive cosmological constraints from results of WP1 and WP2. Our growth rate measurement at low redshift ($z < 0.2$), when combined with high-redshift expansion and growth rate measurements (Fig. 2), has the potential to improve constraints on dark energy and rule out some modified gravity theories [74].

The expansion rate will be provided by SNIa Hubble diagrams, baryon acoustic oscillations (BAO) and the cosmic microwave background (CMB). At high redshifts ($0.2 < z < 2.0$), DESI and Euclid will also provide precise growth rates from redshift-space distortions.

Task 3.1: Implications for LCDM

First we will study the implications of our measurement in a ΛCDM model, where dark energy is considered as a cosmological constant. In this case, we expect to improve constraints on Ω_m and σ_8 when assuming a ΛCDM model, when using only growth-rate measurements.

If combined with CMB or BAO/SN measurements, we do not expect a large improvement on cosmological parameters when adding our low-redshift growth rate measurement given the little freedom in the ΛCDM model. Therefore, obtaining constraints from growth-rate only measurements is of great interest as a consistency check.

We plan to compare our (σ_8, Ω_m) constraints to those from weak-lensing surveys such as Dark Energy Survey (DES, [75]), which show slight tension with those obtained from the CMB [76]. While not decisive, we cannot predict if our measurements will be in more agreement with one or the other.

Task 3.2: Implications for modified gravity

Concerning the $w_0 w_a$CDM model where dark energy is considered to have an evolving equation of state, our measurement of the growth rate, even if at low redshift, does not add much to the already tight constraints from the combination of CMB, BAO and SNIa. However, we are able to put constraints on departures from GR, which could indicate the need for modified gravity theories.

We will first explore the model where $f(z)\ [\Omega_m(z)]^\gamma$, where $\gamma = 0.55$ in general relativity. Measuring departures from this value might point to the need for a modified gravity theory. Figure 4 shows our forecasted constraints on γ, σ_8 and Ω_m when considering only high-redshift growth rate measurements from final DESI samples (red lines) and when adding our measurement from COSMOPV (black lines). We see that a single growth rate measurement has the potential to reduce uncertainties on γ by a factor of ~ 2 (from 0.12 to 0.07 in Figure 4). We also expect an improvement on σ_8 of a few percent.

We will also explore the (μ_0, Σ_0) model which empirically parametrize deviations from general relativity at the level of Poisson equations. This parametrization is interesting when combining growth rate measurements with weak-lensing ones, which are orthogonal in this parameter space. Results from WP2 will particularly useful, since it will be necessary to account for the correlations between growth rate measurements from peculiar velocities and those from weak lensing.

Those results will set the tightest constraints from the Northen sky. This work serve as basis for future analyses of datasets in the Southern sky such as Rubin LSST and the 4-meter Multi-Object Spectroscopic Telescope.

Risk management WP3 requires the completion of WP1 and WP2, therefore there is a **medium level of risk**, encompassing risks of each work pacakge and potential delays in the execution of COSMOPV. Given that this project is a collaborative work, there will be a team of people outside COSMOPV that will help pushing those analyses to completion.

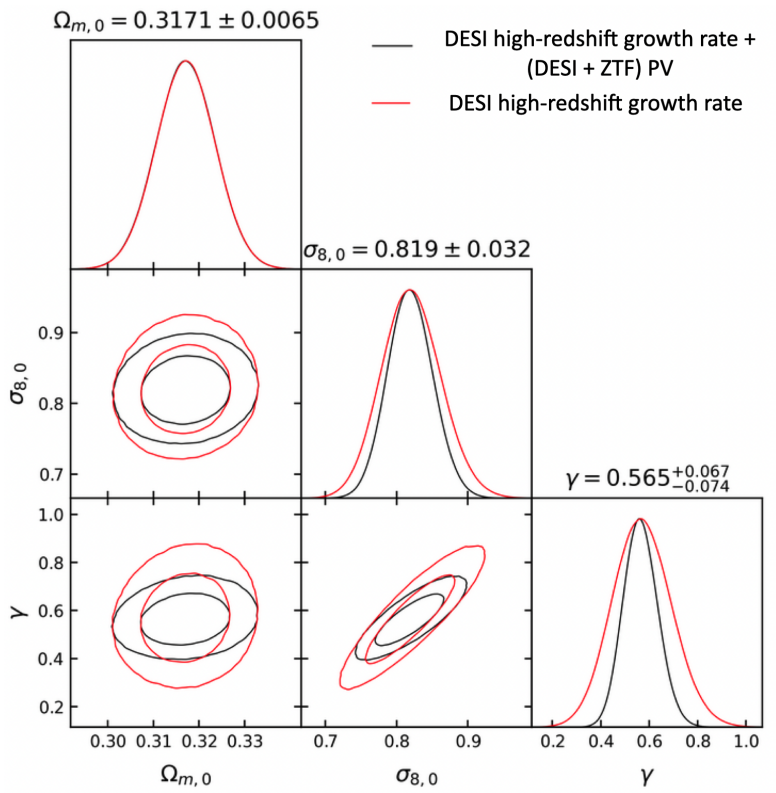

Figure 4. Forecasted constraints on γ, σ_8 and Ω_m from the final DESI growth rate measurements at high-redshift ($z > 0.1$, red lines) and when adding our low-redshift growth rate measurement from DESI and ZTF peculiar velocities (black lines).

II. Organisation and implementation of the project

COSMOPV is a PRME project that will be hosted at Aix-Marseille Université (amU), particularly at the Centre de Physique des Particules de Marseille (CPPM). The team consists of researchers already active in the field, all members of either DESI and ZTF collaborations, spaning a wide range of expertise.

a. The scientific team at the CPPM

1. **Julian Bautista - Scientific Coordinator (Professor, 100% ETPR)**
   Expert in galaxy clustering, type-Ia supernovae and peculiar velocities for cosmology. Major contributor of past cosmological results with spectroscopic surveys such as SDSS-III BOSS, SDSS-IV eBOSS. Currenly active member of DESI and ZTF and actively working in peculiar velocities for the past 5 years, with the support of the Chaire d'Excellence A*Midex. Current co-chair of the DESI Transients and Low-redshift Cosmology Working Group, whose main goal is to perform growth rate measurements with the DESI Peculiar Velocity Survey.
2. **Dominique Fouchez (CNRS Research Director, 10% ETPR)**
   Expert in supernova cosmology (SNLS, ZTF and LSST), photometric detection of sources as well as applications of machine learning to astronomy. Currently PhD supervisor of Rafael Kebadian (see below).
3. **Benjamin Racine (CNRS Researcher, 10% ETPR)**
   Expert in cosmology with the CMB and SNIa. Leading the efforts to homogenize the flux calibration of ZTF images to produce datasets suited for precision cosmology with SNIa. His raw data expertise is of extreme value for COSMOPV, as he will be able to advice on potential sources of systematic effects that impact cosmological results. Currently PhD co-supervisor of Rafael Kebadian (see below).
4. **Fabrice Feinstein (Professor, 40% ETPR)**
   Expert on ZTF datasets. He has been closely working with B. Racine on on the flux calibration of ZTF image and has experience in supernova cosmology. Will contribute with the investigation of observational

systematics in the data (WP 2).

5. **Juan Mena-Fernandez (Postdoc, 30% ETPR)**
   Expert in galaxy clustering, weak lensing and mock catalog production. He is active member of DESI, DES and Rubin-LSST. His expertise will be useful for WP 1 and WP 3.
6. **Corentin Hanser (Postdoc, 30% ETPR)**
   Transitioning from cosmology with galaxy clusters to peculiar velocities. He has been active in setting up a careful blinding procedure for growth rate measurements with SNIa peculiar velocities. He will contribute during one year to the project, in particular to WP1.
7. **Rafael Kebadian (PhD student, 30% ETPR)**
   Currently leading the measurement of the growth rate of structures with half of ZTF data (DR2). He will be part of the project during his last year and will contribute to the first two WP of COSMOPV.

Implication of the scientific coordinator in on-going projects

Name of Researcher	Person. month	Call, funding agency, grant allocated	Title of the project	Name of the scientific coordinator	Start - End
Julian Bautista	N/A	N/A	N/A	N/A	N/A

b. Implemented and requested resources to reach the objectives

Including fees, **the total budget of COSMOPV amounts to 473 697,93 € for a duration of 48 months.** It is divided between staff expenses, travel support, computers, conference fees and conference organisation, plus admin fees.

Staff expenses: 391,3k €

This amount would allow us to hire:

- one PhD student for 36 months for a cost of 137 977,00 €;
- one senior postdoctoral researcher for 36 months for a cost of 253 378,00 €.

The senior postdoctoral researcher would have more than 2 years of experience after their PhD degree.

Equipments: 6k €

Two personnal computers for the future PhD student and postdoctoral researcher, each one at the maximum cost of 3k €. Recently the price of RAM memory increased considerably, driving up the cost of new computers.

Those computers will be essential to the project as they will serve the two new members of the team to develop analysis code and acces the supercomputers (Mardec, NERSC) where the high-performance computing will be performed.

Service delivery: 3 k€

We will publish our findings in the Astronomy & Astrophysics journal (A&A). Recent changes to publication policy limit the number of pages to a maximum of 12 without publication fees. Currently, each extra page costs 150 €. Since our publications often exceed this limit, we will request 3k € for publication fees.

As a reminder, open access versions of our published articles will be posted on the arXiv website without any associated fees.

Other costs: 30,5 k€

We would like to ask **19k € for travel support** for each member of the team, including the scientific coordinator. Each new member of the team will be assisted with 2,5k € per year ($2 \times 3 \times 2,5 = 15k$ €), while the scientific coordinator will be assisted with 1k € per year ($4 \times 1k$ € $= 4k$ €).

We ask an additional **4k € for the payment of conference fees** for the same people for the duration of the project.

To increase the visibility of COSMOPV, as well as our local institutions, we would like request **7,5k € to organise a version of the CosmicFlows conference in Marseille**. Such funds will help with the costs of the conference, such as room rentals, coffee breaks and conference dinner. We expect to welcome around 50 people for a duration of a week, to share about the domain of cosmology and astrophysics with peculiar velocities.

Requested means by item of expenditure and by partner

Type of expenditure	Amount (€)
Staff expenses	391 355,00
Equipments	6 000,00
Service delivery	3 000,00
Other costs	30 500,00
Structure costs**	58 165,43
Sub-total	489 020,43
Requested funding	489 020,43

III. Impact and benefits of the project

a. Scientific impact

Publications: Publication of results in the grade A peer-review journal Astronomy and Astrophysics (A&A). We will ensure that open-access versions of our publications will be published on the arXiv website, as required by the CNRS publication policy.

Conferences: Our results will be advertised in the main cosmological conferences around the world. Presentations are expected by the two new members of the group as well as the scientific coordinator. These presentations will also contribute to the exposure of COSMOPV members as they search their next scientific opportunities.

Organisation of a local CosmicFlows conference: During the project, we plan to organise a version of the CosmicFlows conference locally in Marseille. This conference unites experts from the field of cosmology and astrophysics with peculiar velocities. The scientific coordinator was one of the invited speakers of the last version of the CosmicFlows[3] in 2025 in Brisbane, Australia, tying up good links with their organizers.

b. Societal impact

Press-releases: Major results within the DESI or the ZTF collaborations can be part of a coordinated world-wide press-releases. Those press-releases are usually republished by main news papers such as Le Monde, Le Figaro, among others, significantly increasing the impact of COSMOPV to the general public.

Outreach events: Reaching out to the general public is one of the priorities of COSMOPVṪhe members of the team will be invited to contribute to various outreach events in town and connect to the next generation of potential researchers.

We will continue participating to local outreach events such as *La Fête de la Science*, Pint of Science, as well as promoting our science through workshops for high-school students.

[3] https://indico.global/event/12931/

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